UE22CS441A - Capstone Project Phase-3

Project Progress Review #1

Federated Learning for Privacy-Preserving Medical Image Analysis

Project ID: [Project ID]

Project Guide: [Project Guide Name]

Project Team: [Team Member 1], [Team Member 2], [Team Member 3], [Team Member 4]

Abstract and Scope of the Project

Abstract

- Chest X-ray images (NORMAL vs. PNEUMONIA) are classified using a deep convolutional network trained collaboratively across three simulated clients (hospitals) without ever centralizing raw patient images.
- Federated averaging alone (FedAvg) degrades under non-IID client data; this project uses SCAFFOLD, which adds a control-variate correction to local gradients to counter client-specific drift.
- Baseline result (DenseNet121 backbone, 20 communication rounds): 80.79% accuracy, 80.77% precision, 80.83% recall, 80.80% F1-score on a held-out, class-balanced test set.

Scope

- Binary classification only (NORMAL vs. PNEUMONIA) -- not a multi-disease or multi-label chest X-ray classifier.
- Federated clients are simulated partitions of one public dataset (RSNA Pneumonia Detection Challenge), not a live multi-hospital deployment; each client is deliberately given a different NORMAL:PNEUMONIA ratio (12.5% / 24.4% / 17.7%) to emulate realistic cross-institution heterogeneity.
- Delivered as a research prototype with a Streamlit demonstration app that classifies an uploaded X-ray and generates a downloadable, radiology-style PDF report -- not validated for clinical use.
- Current phase extends the baseline with an EfficientNet-B2 backbone, a class-balanced focal loss tuned per-client, and a corrected (previously incomplete) SCAFFOLD implementation.

Summary of Work Done in Previous Phase

Data Pipeline

- RSNA DICOM images and labels processed with pydicom: MONOCHROME1 polarity correction, intensity normalization, resize to model input size, conversion to 3-channel tensors.
- Training data partitioned into 3 non-IID federated clients and a fully held-out validation + test split.

Federated Training

- SCAFFOLD implemented with Flower + PyTorch on a pretrained DenseNet121 backbone (custom binary head, selective layer unfreezing from denseblock3 onward), dynamic learning-rate schedule, class-weighted sampling, 20 rounds x 2 local epochs x batch size 16.
- Multiple training runs compared: 3 random seeds, a 12k-image baseline variant, a multi-seed ensemble (later removed to keep one authoritative model), and a chest-X-ray-domain-pretrained backbone (TorchXRayVision) experiment that underperformed and was documented as a negative result.

Best Result and Demonstration

- Best model (seed 42, 20 rounds): 80.79% accuracy / 80.77% precision / 80.83% recall / 80.80% F1.
- Streamlit application built: upload an X-ray, get a NORMAL/PNEUMONIA prediction with confidence, and download a structured, radiology-style PDF report.

Inferences Drawn from Literature Survey

- FedAvg (McMahan et al., 2017) averages independently-trained client models, but this compromise worsens as client data heterogeneity increases -- directly relevant given this project's clients have different pneumonia rates.
- SCAFFOLD (Karimireddy et al., 2020) provably reduces this 'client drift' by adding a control-variate correction, (c_global - c_local), to every local gradient step -- adopted as the aggregation algorithm here.
- Naive inverse-class-frequency loss weighting over-corrects for imbalance; Class-Balanced Loss (Cui et al., 2019) instead weights by the 'effective number' of samples, which better matches how much genuinely new information each additional sample provides.
- Focal Loss (Lin et al., 2017) down-weights already-confident predictions so gradient signal concentrates on hard/misclassified examples -- combined with class-balancing for this project's loss function.
- DenseNet121 (Huang et al., 2017) and CheXNet/CheXpert-style transfer learning (Rajpurkar et al., 2017; Irvin et al., 2019) are established strong baselines for chest X-ray classification, motivating the original backbone choice.
- EfficientNet's compound scaling (Tan and Le, 2019) offers a better accuracy-per-parameter trade-off than manually scaled CNNs, motivating the backbone experiment now in progress.
- Overall inference: no single technique (architecture, loss, or aggregation) fully resolves combined non-IID federated drift and class imbalance in medical imaging -- the literature supports combining

List of Tasks/Modules with Individual Contribution

Modules identified in the codebase are listed below. Member assignment, lines of code, and time spent are left blank for the team to fill in.

Task / Module	Team Member	Lines of Code	Time Spent
Data acquisition & preprocess			
Dataset splitting & client parti			
Federated training core / SCAF			
Model architecture (DenseNet			
Loss function design (class-ba			
Evaluation & threshold tuning			
Streamlit demonstration appli			
Documentation & reporting			

Design of Overall Architecture

Data Layer

- RSNA chest X-ray DICOM files and labels -> preprocessing -> 3 non-IID client partitions + a held-out validation set + a held-out test set (touched only once, at final evaluation).

Federated Client Layer (x3)

- Each client holds a local DenseNet121/EfficientNet-B2 model copy, a local control variate (c_i), and trains locally for a fixed number of epochs per round, applying the SCAFFOLD gradient correction (c_global - c_i) at every local step.

Federated Orchestration Layer

- A Flower server strategy aggregates client model updates (sample-weighted averaging) and client control-variate deltas (SCAFFOLD's server-side update) each round, then broadcasts the new global model and global control variate (c) back to all clients.

Evaluation & Application Layer

- Best round selected via held-out validation F1; decision threshold tuned on validation only; final metrics computed once on the untouched test set; deployed through a Streamlit app that runs inference and generates a downloadable PDF report.

Demonstration and Testing of Completed Modules

Preprocessing & Data Modules

- Verified DICOM decoding, MONOCHROME1 polarity handling, and resizing produce correctly oriented, normalized images; model input/output shapes checked (test_model_shapes.py).

Federated Training & Evaluation

- 20-round SCAFFOLD training completed and reproducible across 3 independent seeds; per-round accuracy/precision/recall/F1 logged every round; best-validation-round checkpoint selection verified.
- Threshold selection tested against multiple objectives (accuracy, precision, recall, F1, F-beta); confusion matrix and full classification report generated on the held-out test set.

Application

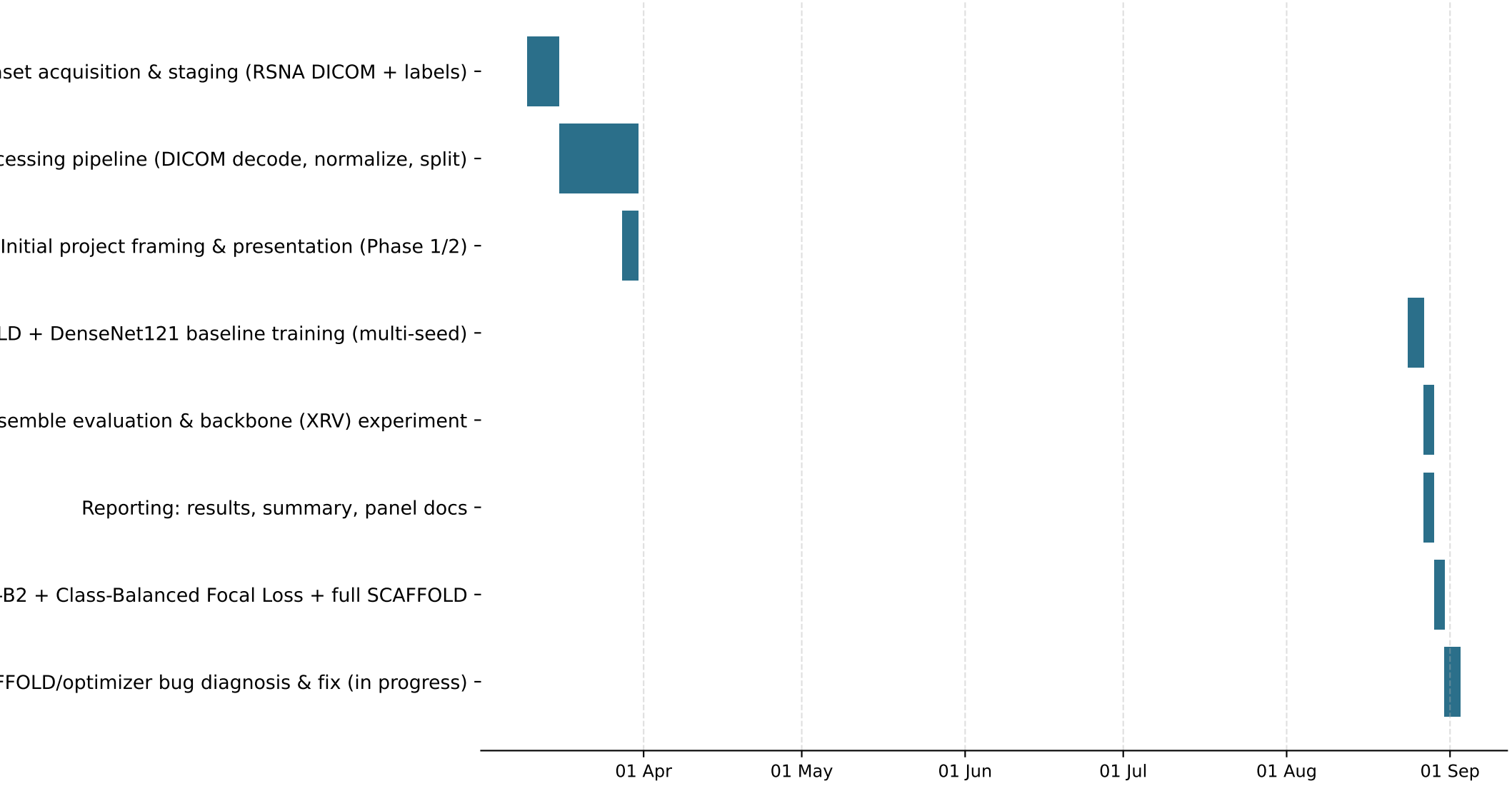
- Streamlit demo manually tested end-to-end: image upload, prediction with confidence score, and PDF report generation and download.

Current Phase (in progress, transparently reported)

- New EfficientNet-B2 + class-balanced focal loss + corrected full-SCAFFOLD pipeline unit-tested via each module's self-test (shape checks, per-client loss sanity checks, threshold-utility validation) and an end-to-end training dry run.
- Testing surfaced a real bug: SCAFFOLD's control-variate math assumes plain SGD, but the pipeline was

Gantt Chart

Timeline reconstructed from actual project file history (dataset staging, experiment logs, report generation).



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Thank You

Questions and feedback are welcome.